

Computer Vision

Activity - 1

1.) Medical Image Enhancement

(i) which gray level transformation would you recommend?

⇒ Log Transformation.

(ii) Why it is suitable?

⇒ Log transformation enhances low-intensity (dark) pixel values while compressing high-intensity values. This improves the visibility of darker tissues without greatly affecting the bright bones.

(iii) What effect will it have on the image?

- Dark tissues become brighter and more visible.
- Bone details remain visible.
- Overall contrast in darker regions improves.

Night - Time Surveillance

(i) Log Transformation.

(ii) The image contains mostly low pixels value (0-40). Log transformation expands these dark intensity values, making the image brighter. Negative transformation only inverts the image and does not effectively improve visibility.

- (iii) • The parking lot appears brighter
• Dark areas reveals more details.

3.)

(i) Power-law (gamma) Transformation

(ii) Power-law transformation increase the contrast b/w fingerprint ridges and valleys making the fine ridges details cleaner and improves fingerprint recognition accuracy.

(iii) Log transformation would brighten the darker regions but may reduce the contrast b/w the ridges and valleys. as a result the fingerprint details would become less distinct, making recognition less effective.